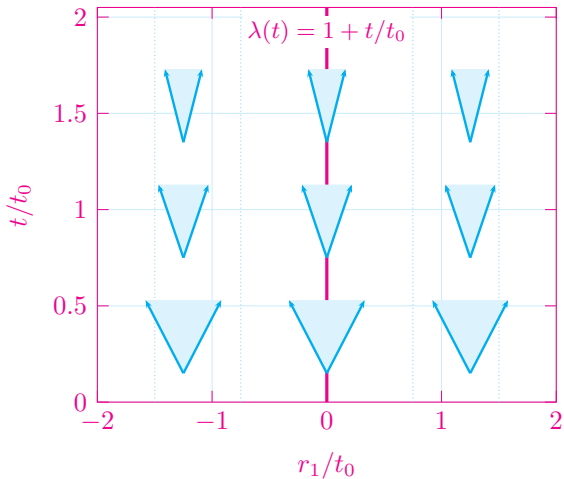
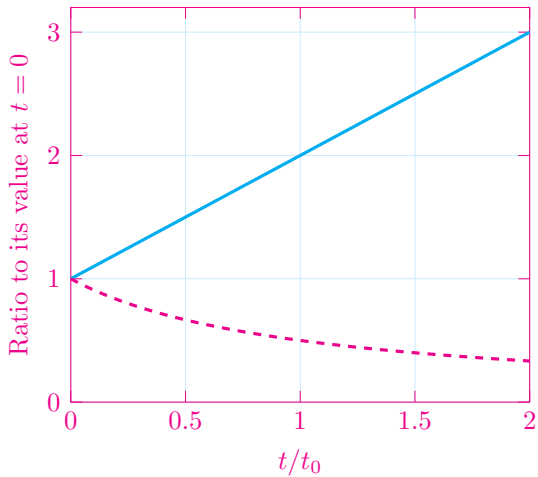


Expansion: local light cones



- Comoving observer
- Local light directions

Separation and momentum



- Separation $\ell/\ell_0 = \lambda$
- Momentum $P/P_0 = 1/\lambda$